

Robustness of Statistical and Machine-Learning Regression Models under Violations of Classical Regression Assumptions: A Simulation Study

Abstract

Regression models are commonly evaluated under assumptions concerning the error structure, predictor relationships, and functional form of the data. When these assumptions are violated, predictive performance may change substantially. This study investigates the behavior of five regression models, Ordinary Least Squares (OLS), Ridge Regression, Huber Regression, Random Forest, and XGBoost, under several controlled departures from classical regression conditions.

A simulation-based framework was used to examine model performance under baseline conditions, heteroscedasticity, outliers, multicollinearity, and nonlinearity. Model performance was evaluated primarily using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 . An additional real-world experiment was conducted using the California Housing dataset.

The results show that model performance depends strongly on the data-generating condition. Under baseline and multicollinearity conditions, OLS, Ridge, and Huber produced very similar results. Under heteroscedasticity, these three models also remained relatively close, while Random Forest and XGBoost produced higher prediction errors in the reported simulation. Under increasing outlier contamination, RMSE increased for all five models. Huber produced slightly lower RMSE than OLS and Ridge at each contamination level, although the differences were modest. In the nonlinear simulation, the linear models performed better than the tree-based models in the reported experiment, demonstrating that the expected advantage of a nonlinear model is not guaranteed under every nonlinear data-generating mechanism. In the California Housing experiment, Random Forest produced the lowest RMSE and MAE and the highest R^2 among the five tested models.

Overall, the findings demonstrate that no single model exhibits uniformly superior performance across all experimental conditions. The study highlights the importance of evaluating regression models under multiple forms of assumption violation rather than relying on performance under a single data-generating scenario.

1. Introduction

Regression analysis is one of the fundamental tools used in statistics and machine learning for explaining relationships between variables and making predictions. Classical linear regression provides a well-established framework for estimating relationships between predictors and a response variable. However, the validity and interpretation of classical regression methods can be affected when important assumptions are violated.

Common problems include heteroscedasticity, the presence of outliers, multicollinearity among predictors, and nonlinear relationships between predictors and the response. These conditions may affect estimation, prediction accuracy, coefficient stability, and the reliability of statistical conclusions.

Machine-learning methods provide alternative approaches to regression modelling. Tree-based methods such as Random Forest and XGBoost can model nonlinear relationships and interactions without requiring the same functional form as a conventional linear regression model. Robust regression methods such as Huber Regression are designed to reduce the influence of extreme observations. Ridge Regression introduces regularization that can help address problems associated with correlated predictors.

However, the performance of these approaches depends on the characteristics of the data. A model that performs well under one form of model misspecification may not necessarily perform well under another. Therefore, systematic comparison under controlled experimental conditions is useful for understanding the strengths and limitations of different approaches.

This study extends a statistical perspective on regression-assumption violations into a broader comparison involving both classical statistical models and machine-learning methods.

2. Research Question

The main research question is:

How do different statistical and machine-learning regression models behave when classical regression assumptions are progressively violated, and how does their predictive performance change across different data-generating conditions?

The study compares:

1. Ordinary Least Squares (OLS)
2. Ridge Regression
3. Huber Regression
4. Random Forest Regression
5. XGBoost Regression

3. Objectives

The objectives of the study are:

6. To establish a baseline comparison of five regression models under a controlled linear data-generating process.
7. To examine model performance under heteroscedasticity.
8. To investigate the effect of increasing response-outlier contamination.
9. To evaluate model performance under multicollinearity.
10. To examine model behavior when the true relationship contains nonlinearity.
11. To compare the simulation findings with performance on a real-world regression dataset.
12. To assess whether model performance remains consistent across different forms of assumption violation.

4. Models Considered

4.1 Ordinary Least Squares

OLS is the classical linear regression method used as the primary statistical benchmark. It estimates regression coefficients by minimizing the sum of squared residuals.

4.2 Ridge Regression

Ridge Regression extends linear regression by adding an L2 penalty to the objective function. It can reduce the influence of highly correlated predictors and stabilize coefficient estimates.

4.3 Huber Regression

Huber Regression is a robust regression method that uses a loss function that behaves quadratically for small residuals and more linearly for large residuals. This reduces the influence of extreme residuals compared with ordinary squared-error loss.

4.4 Random Forest

Random Forest is an ensemble tree-based method that combines predictions from multiple decision trees. It can capture nonlinear relationships and interactions between predictors.

4.5 XGBoost

XGBoost is a gradient-boosting method that constructs an ensemble of trees sequentially. It is capable of representing nonlinear relationships and complex interactions.

5. Methodology

5.1 Simulation Framework

Controlled data-generating processes were used to investigate different departures from classical regression assumptions.

For the baseline experiment, the response variable was generated according to a linear relationship:

$$Y = 3 + 2X_1 + 1.5X_2 - X_3 + \varepsilon$$

where the predictors were generated from standard normal distributions and the error term followed a normal distribution.

The experimental conditions were then modified to introduce specific forms of model misspecification.

5.2 Heteroscedasticity

For the heteroscedasticity experiment, the standard deviation of the error term was allowed to depend on the magnitude of X_1 :

$$\sigma(X_1) = 0.5 + 1.5|X_1|$$

This creates a situation in which the variability of the response changes across observations.

5.3 Outlier Contamination

Response outliers were introduced at four contamination levels:

- 0%
- 2%
- 5%
- 10%

The outlier values were generated by adding large random disturbances to selected response observations.

5.4 Multicollinearity

Multicollinearity was introduced by generating X_2 as a noisy version of X_1 :

$$X_2 = X_1 + \eta$$

where η is a small random error.

This creates strong correlation between predictors.

5.5 Nonlinearity

Nonlinearity was introduced by replacing the linear effect of X_1 with a quadratic term:

$$Y = 3 + 2X_1^2 + 1.5X_2 - X_3 + \varepsilon$$

This allows the study to examine model performance when the true relationship is not purely linear.

5.6 Evaluation Metrics

Three predictive performance measures were used:

RMSE

RMSE measures the square root of the average squared prediction error and gives greater weight to large errors.

MAE

MAE measures the average absolute prediction error and provides a more direct measure of average prediction deviation.

R²

R² measures the proportion of variation in the response explained by the predictions.

For repeated simulations, the mean and standard deviation of the performance metrics were used to assess both average performance and variability.

6. Results

6.1 Baseline Simulation

The baseline experiment was conducted under the controlled linear data-generating process.

Model	RMSE	MAE	R ²
OLS	1.0074	0.8038	0.8754
Ridge	1.0074	0.8038	0.8754
Huber	1.0077	0.8041	0.8754
Random Forest	1.1615	0.9223	0.8345
XGBoost	1.1017	0.8778	0.8510

The baseline results show very similar performance among OLS, Ridge, and Huber. Random Forest and XGBoost produced higher RMSE and MAE and lower R² under the reported baseline setting.

The repeated-simulation results also showed relatively small variation for the three regression models, with the standard deviations of RMSE, MAE, and R^2 remaining close across OLS, Ridge, and Huber.

These results are consistent with the fact that the data-generating relationship is explicitly linear. The additional flexibility of the tree-based models does not necessarily translate into improved predictive performance when the underlying relationship is already well represented by a linear specification.

6.2 Heteroscedasticity

The reported heteroscedasticity results were:

Model	RMSE	MAE	R^2
OLS	1.786699	1.332272	0.701158
Ridge	1.787095	1.332826	0.701025
Huber	1.783257	1.328901	0.702308
Random Forest	2.017521	1.482035	0.618956
XGBoost	1.936975	1.424232	0.648774

OLS, Ridge, and Huber remained relatively close to one another. Huber had the lowest RMSE and MAE among the five models in this particular experiment, although the differences from OLS and Ridge were small.

Random Forest produced the highest RMSE and MAE and the lowest R^2 in the reported heteroscedasticity experiment. XGBoost performed between the linear/robust regression models and Random Forest.

These results indicate that the presence of heteroscedasticity did not automatically give the flexible tree-based models an advantage in this simulation.

6.3 Outlier Contamination

The effect of increasing response-outlier contamination was evaluated using RMSE.

Outliers	OLS	Ridge	Huber	Random Forest	XGBoost
0%	1.046290	1.046338	1.046298	1.214764	1.145818
2%	2.434032	2.434003	2.425874	2.611366	2.573664
5%	3.266774	3.266738	3.251857	3.451933	3.405317
10%	4.749851	4.749860	4.730177	5.015014	4.940420

A clear increase in RMSE occurred for all five models as the proportion of outliers increased.

Huber produced the lowest RMSE at each contamination level in the reported experiment. Its advantage over OLS and Ridge, however, was relatively small. At 10% contamination, for example, the RMSE values were 4.7302 for Huber, 4.7499 for OLS, and 4.7499 for Ridge.

Random Forest had the highest RMSE at every reported contamination level. XGBoost also experienced substantial deterioration as contamination increased.

The results therefore indicate that increasing response contamination substantially affects predictive accuracy for all five models. Huber showed a modest advantage under this particular outlier-generation mechanism, but the results do not support the conclusion that it is completely insensitive to outliers.

6.4 Multicollinearity

The reported multicollinearity results were:

Model	RMSE	MAE	R ²
OLS	1.036938	0.827121	0.921284
Ridge	1.034399	0.824587	0.921669
Huber	1.037427	0.828088	0.921210
Random Forest	1.199706	0.981640	0.894633
XGBoost	1.138818	0.891024	0.905057

The three regression models produced very similar results. Ridge had the lowest RMSE and MAE and the highest R² among the five models in the reported experiment, although its differences from OLS and Huber were very small.

Random Forest and XGBoost had higher RMSE and MAE and lower R².

The results suggest that the particular degree of multicollinearity introduced in the simulation did not substantially reduce the predictive performance of the linear models. It is important to distinguish predictive performance from coefficient stability, however. Multicollinearity can have important consequences for coefficient estimation and interpretation even when prediction error changes only slightly.

6.5 Nonlinearity

The reported nonlinear simulation results were:

Model	RMSE	MAE	R ²
OLS	1.036938	0.827121	0.876029
Ridge	1.037339	0.827542	0.875933
Huber	1.037429	0.828090	0.875911
Random Forest	1.228379	0.956959	0.826028

Model	RMSE	MAE	R ²
XGBoost	1.132448	0.897738	0.852140

An important finding is that the tree-based models did not outperform the linear models in this particular nonlinear simulation.

OLS had the lowest RMSE and MAE and the highest R² among the five models, followed closely by Ridge and Huber. XGBoost performed better than Random Forest but remained below the three regression models on the reported metrics.

This result demonstrates why model comparison should be based on empirical experiments rather than assumptions about which class of model should perform best. The specific form and strength of the nonlinearity, sample size, noise level, and model configuration can all influence the outcome.

6.6 California Housing Application

The final experiment applied the five models to the California Housing dataset.

Model	RMSE	MAE	R ²
OLS	69,297.72	50,413.43	0.6488
Ridge	69,250.31	50,391.14	0.6493
Huber	76,534.64	55,733.46	0.5717
Random Forest	48,757.92	31,663.84	0.8262
XGBoost	56,343.41	38,938.38	0.7679

The real-world application produced a substantially different pattern from several of the controlled simulations.

Random Forest had the lowest RMSE and MAE and the highest R² among the five models in this experiment. XGBoost also performed better than the three linear/robust regression models on the reported predictive metrics.

OLS and Ridge produced very similar results, while Huber produced the highest RMSE and MAE and the lowest R² among the five models.

The difference between the simulation results and the California Housing results illustrates the importance of data structure. Real-world datasets may contain nonlinearities, interactions, heterogeneous relationships, and other features that are not fully represented by a simple controlled simulation.

7. Overall Comparison

The experiments demonstrate that model performance is condition-dependent.

OLS, Ridge, and Huber were generally close under the controlled linear, heteroscedastic, and multicollinear settings. Their relative differences were often small.

Ridge showed a particularly similar predictive profile to OLS in several experiments, which is consistent with the fact that the simulated predictors were not associated with an extremely severe multicollinearity problem.

Huber showed a modest advantage under the specific response-outlier contamination mechanism, producing the lowest RMSE at all four outlier levels. However, its performance still deteriorated substantially as contamination increased.

Random Forest and XGBoost did not consistently outperform the statistical regression models in the controlled simulations. In particular, Random Forest produced higher errors in the reported heteroscedasticity, multicollinearity, and nonlinearity experiments.

In contrast, the California Housing experiment showed a substantially different pattern. Random Forest and XGBoost produced lower prediction errors and higher R^2 than the three regression models.

Therefore, the results do not identify one universally superior model. Instead, they show that model performance depends on the underlying data-generating process and the type of deviation from classical assumptions.

8. Discussion

One of the central findings of this study is that predictive performance cannot be inferred solely from a model's flexibility or robustness label.

The controlled simulations provide a useful environment for isolating individual phenomena. Under the baseline linear model, the three regression methods were closely matched because the true relationship was linear. Introducing heteroscedasticity increased prediction error, but did not fundamentally change the relative ordering of the models in the reported experiment.

The outlier experiment provided stronger evidence of progressive degradation. All models showed increasing RMSE as contamination increased. Huber had a small advantage over OLS and Ridge, which is consistent with its robust loss function. Nevertheless, the improvement was not large enough to indicate immunity to contamination.

The multicollinearity experiment showed another important distinction. Ridge is designed to address instability associated with correlated predictors, but the predictive differences among OLS, Ridge, and Huber were small in this experiment. This suggests that the degree of multicollinearity introduced was not sufficiently severe to produce a large difference in predictive performance. A future study could examine stronger levels of predictor correlation and evaluate coefficient variability in addition to prediction error.

The nonlinear simulation produced an unexpected but informative result. The tree-based methods did not outperform the linear models. This may reflect the specific quadratic data-generating process, sample size, noise level, train-test split, and hyperparameter settings used in the experiment. It emphasizes that "nonlinear data" does not automatically imply that a particular machine-learning model will produce lower prediction error.

The California Housing application provided a useful real-world contrast. The tree-based models performed substantially better than the linear models on the reported test set. This suggests that the structure of the real-world dataset contains predictive patterns that were better captured by the tree-based approaches under the selected model configurations.

Taken together, these findings support the use of controlled simulation studies alongside real-world applications. Simulation makes it possible to understand model behavior under known conditions, while real-world data demonstrate how those conclusions translate to a complex empirical setting.

9. Limitations

Several limitations should be considered when interpreting the results.

First, the simulations represent specific data-generating mechanisms. Other forms or strengths of heteroscedasticity, outlier contamination, multicollinearity, and nonlinearity could produce different results.

Second, the models were evaluated using particular hyperparameter settings. Different tuning procedures or hyperparameter values could change the predictive performance of Random Forest and XGBoost and could also affect the other models.

Third, the nonlinear experiment used a specific quadratic relationship. Other nonlinear structures, including interactions, thresholds, exponential relationships, or more complex functional forms, could lead to different model comparisons.

Fourth, the outlier experiment focused on RMSE across contamination levels in the reported results. RMSE is especially sensitive to large errors, so complementary robustness analyses using MAE and other measures could provide additional information.

Fifth, not all of the supplied heteroscedasticity, multicollinearity, and nonlinearity results are repeated-simulation mean and standard-deviation summaries. They are reported as the results of the corresponding experiment. Therefore, they should not be interpreted as estimates of sampling variability across 100 independent simulations.

Sixth, the California Housing results are based on one train-test evaluation. Repeated cross-validation or repeated train-test splits would provide a stronger assessment of generalization performance.

Finally, the study focuses primarily on predictive metrics. For statistical regression, coefficient stability, confidence intervals, inference, and interpretability are also important considerations.

10. Future Research

Future work could extend this study in several directions.

1. Increase the number of repeated simulations for every assumption-violation experiment and report mean $\pm$ SD for all models.
2. Examine multiple levels of heteroscedasticity rather than a single error-variance structure.
3. Evaluate several types and magnitudes of outlier contamination.
4. Increase the severity of multicollinearity and examine coefficient stability and variance inflation.
5. Test multiple nonlinear functions and predictor interactions.
6. Tune Random Forest and XGBoost using cross-validation.
7. Include additional models such as Elastic Net, Support Vector Regression, Gradient Boosting, and generalized additive models.
8. Compare predictive accuracy with computational cost and model interpretability.
9. Use repeated cross-validation on the California Housing dataset.
10. Investigate uncertainty quantification and prediction intervals under model misspecification.

11. Conclusion

This study compared OLS, Ridge, Huber, Random Forest, and XGBoost under several controlled violations of classical regression assumptions and on a real-world housing dataset.

The results show that no single model performed uniformly better across all experimental settings. OLS, Ridge, and Huber were often closely matched in the controlled simulations. Huber showed a modest advantage under the particular outlier contamination mechanism used, but all models experienced substantial deterioration as contamination increased. The nonlinear simulation also demonstrated that nonlinear machine-learning models do not necessarily outperform linear models under every nonlinear data-generating process.

The California Housing experiment produced a different pattern, with Random Forest and XGBoost achieving lower prediction errors and higher R^2 than the linear and robust regression models under the reported evaluation.

The main implication is that regression-model selection should be informed by the structure of the data and the type of model violation being considered. Simulation studies provide a useful framework for investigating these effects systematically, while real-world validation is necessary to determine how model behavior translates to practical prediction problems.

Overall, this study provides a reproducible framework for examining the robustness and predictive behavior of statistical and machine-learning regression methods under controlled departures from classical assumptions.